

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-3

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

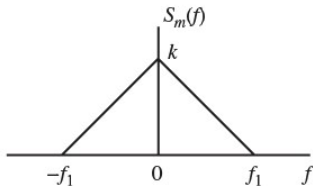
Date: 20-08-2025

Objectives

- Spectral Density, Hilbert filter, Analytical Signal.

Tutorial Problems

1. A signal $m(t)$ with power spectral density $S_m(f)$ as given in the figure is passed through a transmitter block $2f_1 \text{sinc}(2f_1\pi t)$ and then a Hilbert filter with an impulse response $\frac{1}{\pi t}$. Find the power at the output of the transmitter block and comment on the loss of the power by the transmitter block.



2. Determine the analytical signal (or complex envelope) of the signal $x(t) = \cos 2\pi f_1 t + \cos 2\pi f_2 t$, where f_1 and f_2 are positive.

Solution

1. PSD from the figure can be represented as

$$S_m(f) = \begin{cases} k \left(1 - \frac{|f|}{f_1} \right), & |f| \leq f_1, \\ 0, & \text{otherwise.} \end{cases}$$

Input power $P_m = \int_{-\infty}^{\infty} S_m(f) df = 2k \int_0^{f_1} \left(1 - \frac{f}{f_1} \right) df = 2k \left(f_1 - \frac{f_1}{2} \right) = kf_1$.

Alternatively, P_m : Area of PSD = $2 \frac{1}{2} k f_1 = kf_1$.

$$S_y(f) = |H_{LP}(f)|^2 |H_H(f)|^2 S_m(f)$$

where $|H_{LP}(f)|$ and $|H_H(f)|$ are frequency response of LPF and Hilbert filter. Using the property of Hilbert filter, $|H_H(f)|^2 = 1$ and $|H_{LP}(f)| = \text{rect}(\frac{f}{2f_1})$. Thus,

$$S_y(f) = S_m(f)$$

The output power is

$$P_y = P_m = kf_1$$

Comment: There is no loss of power.

2. The analytic signal or complex envelope of the signal is

$$x_a(t) = x(t) + jx_h(t)$$

$$x(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t), \quad f_1, f_2 > 0,$$

$$x_h(t) = \cos(2\pi f_1 t - \pi/2) + \cos(2\pi f_2 t - \pi/2)$$

$$x_h(t) = \sin(2\pi f_1 t) + \sin(2\pi f_2 t)$$

$$x_a(t) = x(t) + jx_h(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t) + j \sin(2\pi f_1 t) + j \sin(2\pi f_2 t) = e^{j2\pi f_1 t} + e^{j2\pi f_2 t}$$

Fourier transform:

$$X(f) = \frac{1}{2} [\delta(f - f_1) + \delta(f + f_1) + \delta(f - f_2) + \delta(f + f_2)].$$

$$X_a(f) = \delta(f - f_1) + \delta(f - f_2), \quad X_a(f) = 0 \text{ for } f < 0.$$

Main take way: Real-valued signals contain both negative and positive spectra but analytical signal has either positive/negative spectra.